

# STATISTICS

1. For the following distribution

|      |      |       |       |       |       |
|------|------|-------|-------|-------|-------|
| C.I. | 0-10 | 10-20 | 20-30 | 30-40 | 40-50 |
| f    | 20   | 30    | 24    | 40    | 18    |

the lower limit of the modal class is

- (a) 20                      (b) 30                      (c) 40                      (d) 50

2. For the following distribution

|      |     |      |       |       |       |
|------|-----|------|-------|-------|-------|
| C.I. | 0-5 | 6-11 | 12-17 | 18-23 | 24-29 |
| f    | 26  | 20   | 30    | 16    | 22    |

the upper limit of the median class is

- (a) 18.5                      (b) 18                      (c) 17.5                      (d) 17

3. For the following distribution the sum of the upper limit and the lower limit of the modal class is

|    |     |      |       |       |       |
|----|-----|------|-------|-------|-------|
| CI | 0-5 | 5-10 | 10-15 | 15-20 | 20-25 |
| f  | 10  | 15   | 12    | 20    | 9     |

- (a) 0                      (b) 5                      (c) 10                      (d) 35

4. The lower limit of the median class of the distribution is \_\_\_\_

|           |        |         |         |          |           |
|-----------|--------|---------|---------|----------|-----------|
| Classes   | 0 – 25 | 25 – 50 | 50 – 75 | 75 – 100 | 100 – 125 |
| Frequency | 18     | 12      | 10      | 15       | 5         |

- (a) 25                      (b) 50                      (c) 75                      (d) 100

5. For the following data the lower limit of the modal class

|                |               |               |               |               |               |
|----------------|---------------|---------------|---------------|---------------|---------------|
| Height (In cm) | Less than 120 | Less than 140 | Less than 160 | Less than 180 | Less than 200 |
| students       | 12            | 26            | 34            | 40            | 50            |

- (a) 120                      (b) 140                      (c) 160                      (d) 180

6. In a continuous frequency distribution if  $n=50$  lower limit of median class is 24.5,  $cf=9$ , frequency of median class is 8 and class size is 10, then what is its median.

- (a) 24.5                      (b) 36.5                      (c) 44.5                      (d) 55

7. The modal class of the following distribution is

|                    |      |       |       |       |        |
|--------------------|------|-------|-------|-------|--------|
| Marks obtained     | 0-20 | 20-40 | 40-60 | 60-80 | 80-100 |
| Number of students | 15   | 18    | 21    | 29    | 17     |

- a) 0 – 20                      (b) 20 – 40                      (c) 40 – 60                      (d) 60 – 80

8. In the following distribution the values of  $x + y$ .

|     |        |         |         |         |         |         |       |
|-----|--------|---------|---------|---------|---------|---------|-------|
| C I | 0 – 10 | 10 – 20 | 20 – 30 | 30 – 40 | 40 – 50 | 50 – 60 | Total |
| f   | 5      | x       | 20      | 15      | y       | 5       | 60    |

9. Find the value of  $x$  if mean of the data given is 27.

|     |        |         |         |         |         |
|-----|--------|---------|---------|---------|---------|
| C I | 0 – 10 | 10 – 20 | 20 – 30 | 30 – 40 | 40 – 50 |
| f   | 8      | x       | 12      | 13      | 10      |

10. Find the mode of the following distribution of marks obtained by the students in an

|                    |      |       |       |       |        |
|--------------------|------|-------|-------|-------|--------|
| Marks obtained     | 0-20 | 20-40 | 40-60 | 60-80 | 80-100 |
| Number of students | 15   | 18    | 21    | 29    | 17     |

examination:

11. Find the median and mode of the data

|                 |       |       |       |       |       |       |
|-----------------|-------|-------|-------|-------|-------|-------|
| Percentage      | 30-35 | 35-40 | 40-45 | 45-50 | 50-55 | 55-60 |
| No. of students | 15    | 16    | 19    | 23    | 18    | 9     |

12. Find the modal monthly pocket money of a child for the following data.

|                                         |         |         |          |           |           |
|-----------------------------------------|---------|---------|----------|-----------|-----------|
| Monthly Pocket money given ( in Rupees) | 30 – 60 | 60 – 90 | 90 – 120 | 120 – 150 | 150 – 180 |
|-----------------------------------------|---------|---------|----------|-----------|-----------|

|                 |    |    |    |    |    |
|-----------------|----|----|----|----|----|
| No. of children | 15 | 24 | 28 | 22 | 12 |
|-----------------|----|----|----|----|----|

13. Median of the following distribution is 28.5. Find the values of  $x$  and  $y$ .

|     |        |         |         |         |         |         |       |
|-----|--------|---------|---------|---------|---------|---------|-------|
| C I | 0 – 10 | 10 – 20 | 20 – 30 | 30 – 40 | 40 – 50 | 50 – 60 | Total |
| f   | 5      | x       | 20      | 15      | y       | 5       | 60    |

14. Median of the following distribution is 525. Find the values of  $x$  and  $y$  if the total frequency is 100.

|                |       |         |         |         |         |         |         |         |         |          |
|----------------|-------|---------|---------|---------|---------|---------|---------|---------|---------|----------|
| Class Interval | 0-100 | 100-200 | 200-300 | 300-400 | 400-500 | 500-600 | 600-700 | 700-800 | 800-900 | 900-1000 |
| Frequency      | 2     | 5       | x       | 12      | 17      | 20      | y       | 9       | 7       | 4        |

15. Find the missing frequency if the mode of the following distribution is 154.

|           |           |           |           |           |           |           |
|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| Classes   | 120 – 130 | 130 – 140 | 140 – 150 | 150 – 160 | 160 – 170 | 170 – 180 |
| frequency | 2         | 8         | 12        | F         | 8         | 7         |

16. The median of the following data is 50. Find the values of 'p' and 'q', if the sum of all frequencies is 90. Also find the mode.

|          |         |         |         |         |         |         |         |
|----------|---------|---------|---------|---------|---------|---------|---------|
| Marks    | 20 – 30 | 30 – 40 | 40 – 50 | 50 – 60 | 60 – 70 | 70 – 80 | 80 – 90 |
| Students | p       | 15      | 25      | 20      | q       | 8       | 10      |

17. The following table gives the distribution of the life time of 400 neon lamps. Find the average life time of a lamp.

|                       |             |             |             |             |             |             |             |
|-----------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Life time ( in hours) | 1500 – 2000 | 2000 – 2500 | 2500 – 3000 | 3000 – 3500 | 3500 – 4000 | 4000 – 4500 | 4500 – 5000 |
| No. of lamps          | 14          | 56          | 60          | 86          | 74          | 62          | 48          |

18. Find the mode of the data whose mean is 20 and median is 22.